

Synoptic CB: Macrophyte Fluxes

2023 Samples

2026-02-23

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#Read in and collate all the files in the Processed Data folder

```
ch4 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2023_CH4.csv")
ch4$X <- NULL
ch4 <- ch4 %>%
  rename_with(
    .cols = 7:(ncol(.) - 3),
    .fn = ~paste0("ch4_", .)
  )
```

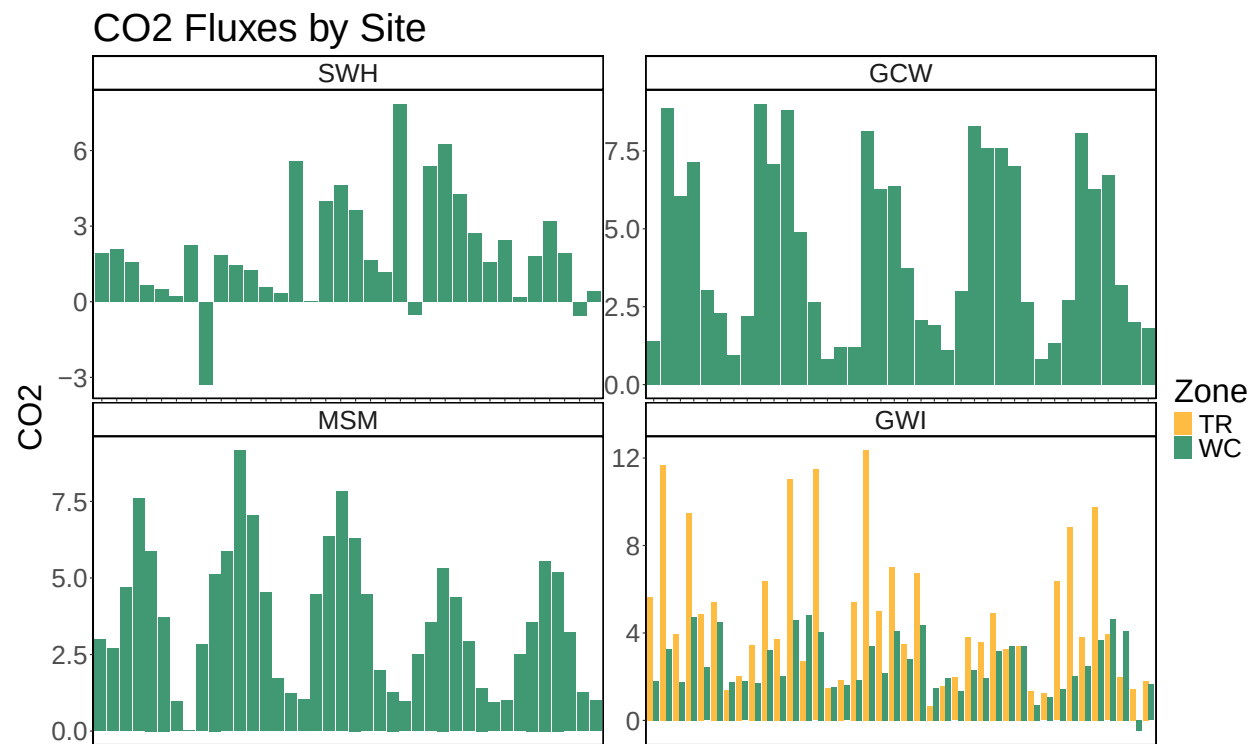
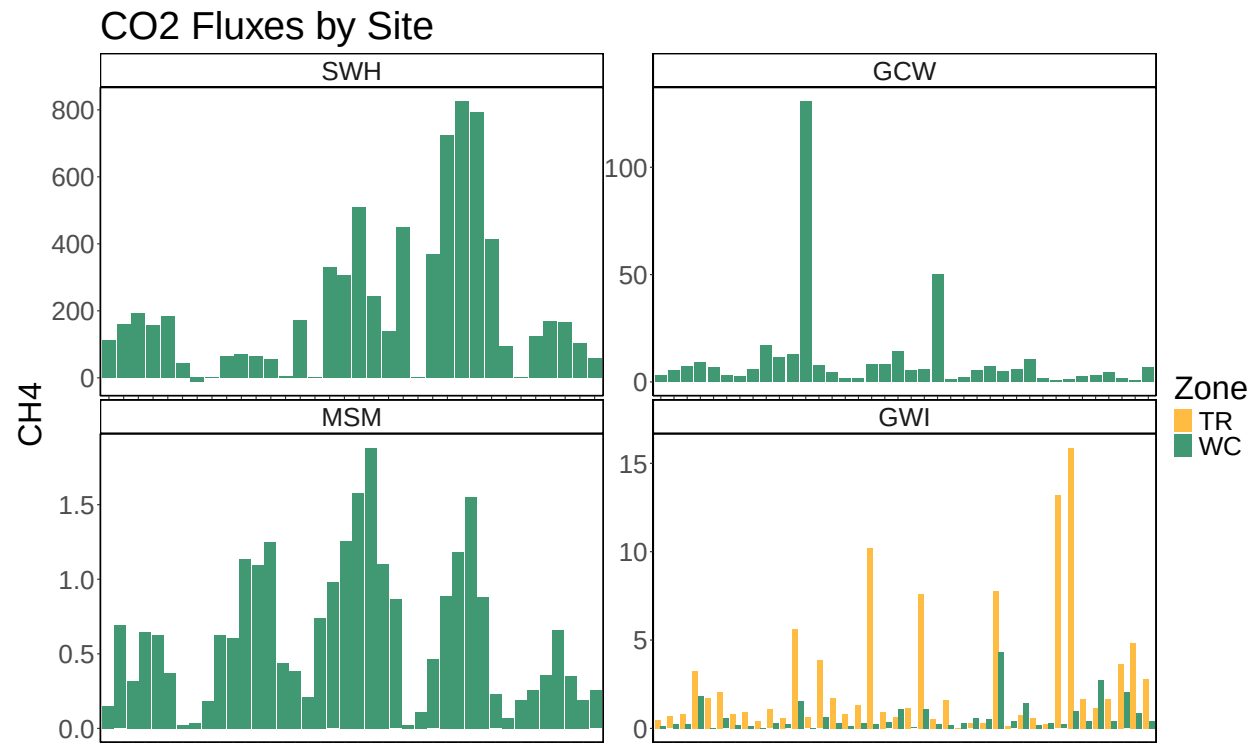
```
co2 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2023_CO2.csv")
co2$X <- NULL
co2 <- co2 %>%
  rename_with(
    .cols = 7:(ncol(.) - 3),
    .fn = ~paste0("co2_", .)
  )
```

Combine all files

```
all_dat <- merge(ch4, co2, by = c("Plot", "Site", "Zone", "Replicate", "Date", "TIMESTAMP", "TIMESTAMP_1"))
```

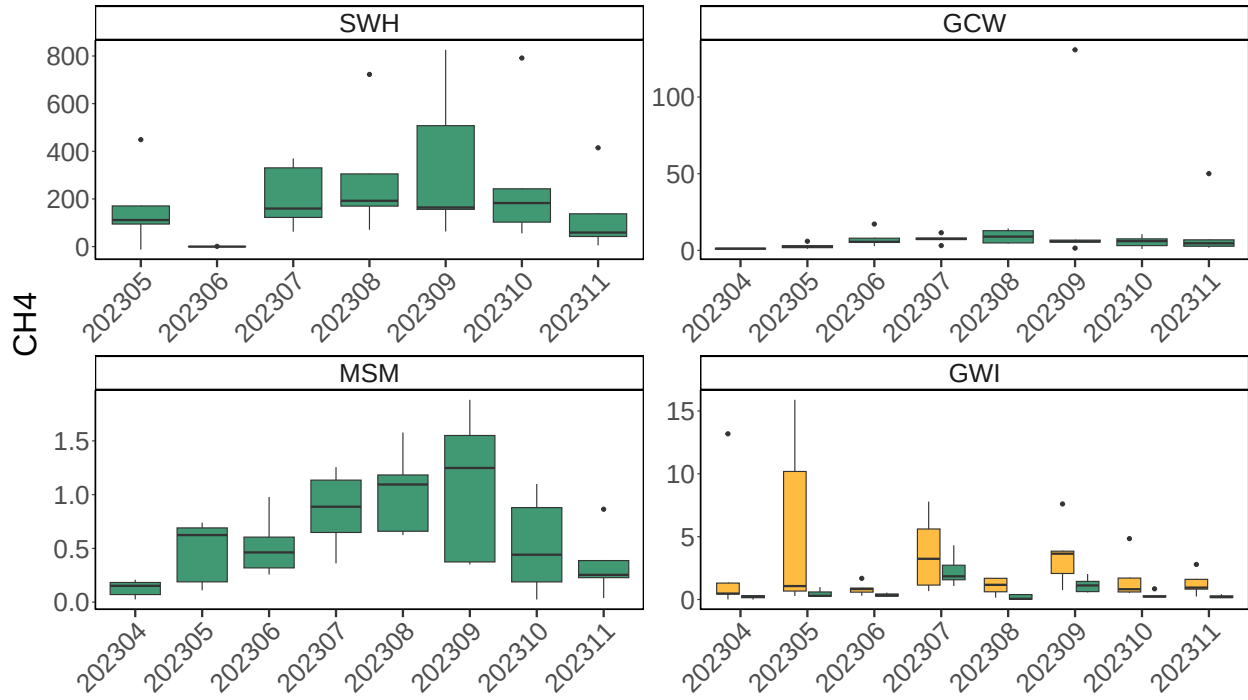
0.1 Write out files

0.2 All Data by Year, Site and Zone

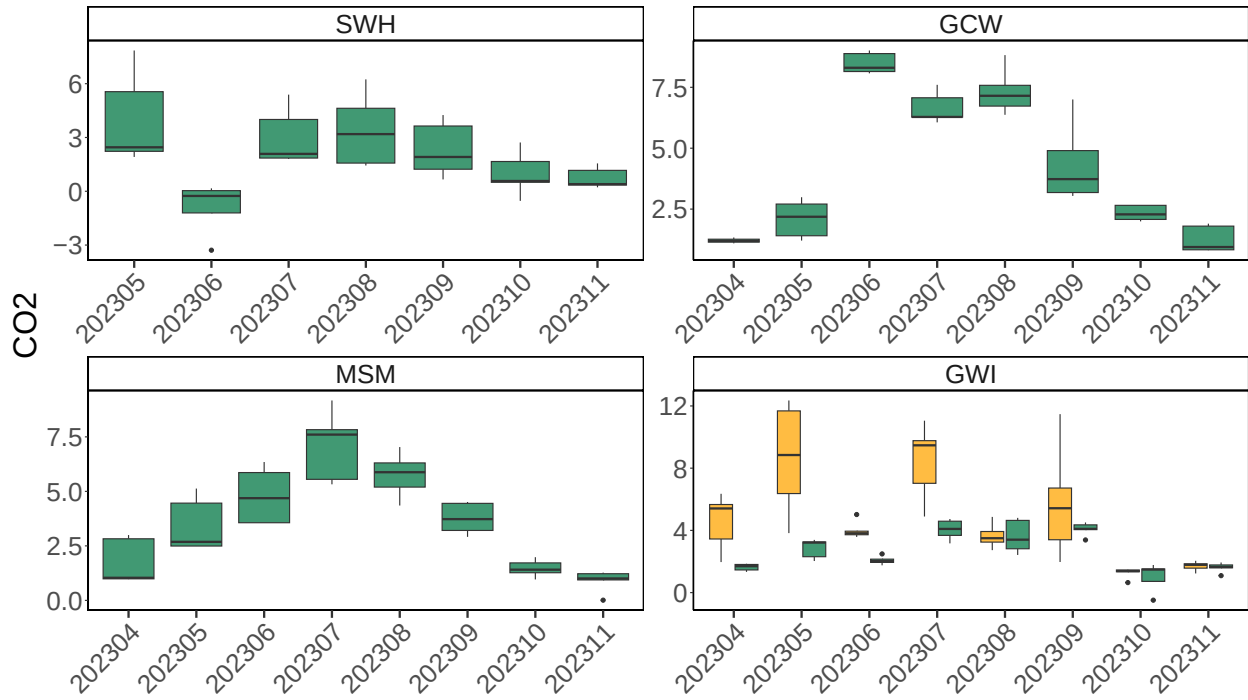


0.3 Boxplot summary by Date

Summary Boxplot of CH4 by Site and Zone

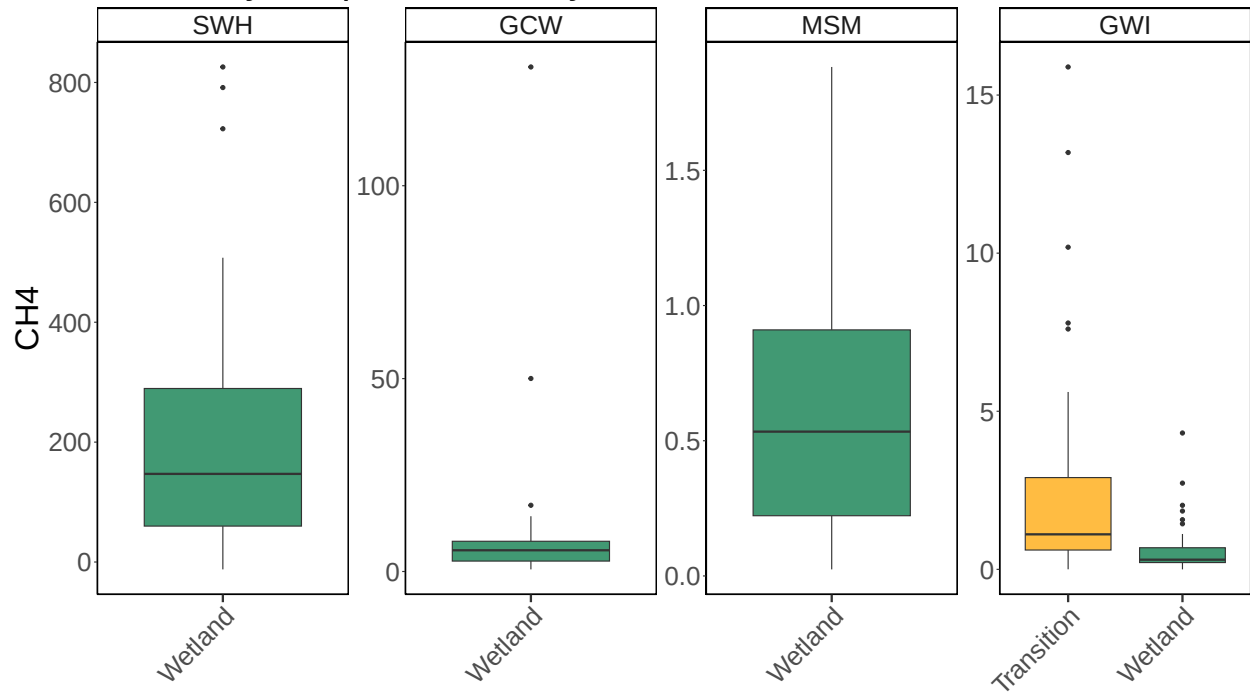


Summary Boxplot of CO2 by Site and Zone

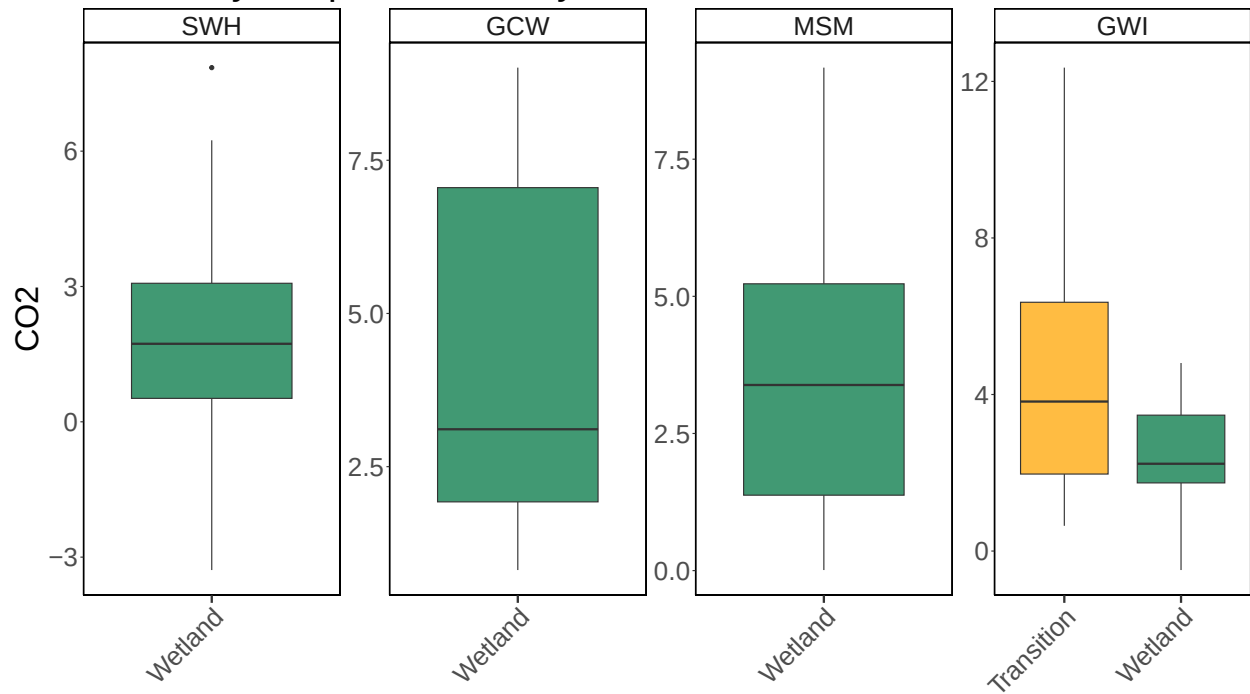


0.4 Boxplot summary by Site

Summary Boxplot of CH₄ by Site and Zone



Summary Boxplot of CO₂ by Site and Zone



##Calculate summary metrics

0.5 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
SWH	Wetland	-12.3011582	825.834316	207.9958785	-3.2864193	7.851624	2.025091
GCW	Wetland	0.5469638	130.778376	10.0769700	0.8129533	9.011439	4.376092
MSM	Wetland	0.0242469	1.882217	0.6210485	0.0101858	9.169040	3.623640
GWI	Transition	0.0020981	15.888051	2.6101187	0.6444756	12.352160	4.756295
GWI	Wetland	-0.0009012	4.312324	0.6584696	-0.4857045	4.804592	2.613157